

Optimal Control of Online Game Addiction: An SEIR Model

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- Online game addiction is considered a growing social problem.
- The World Health Organization (WHO) has classified game addiction as a mental health disorder.
- SEIR model typically used for infectious disease.
- Analyze and control online game addiction via guidance and counseling.

SEIR Model

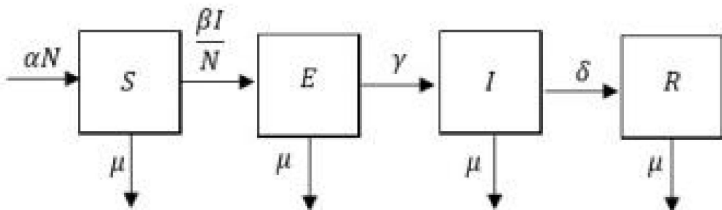


Figure: SEIR Model Schematic for Online Game Addiction

Definition of the Variables and Parameters

Vars & Params	Definition
αN	Proportion of total student population
S	Group of students who are prone to online game addiction
E	Group of students who play online games
I	Group of students who are addicted to online games
R	Group of students who are not addicted to online games
β	The rate at which students who are prone to online game addiction (S) start to play online games (E)
γ	The rate at which students who play online games (E) become addicted (I)
δ	The rate at which students who are addicted to online games (I) overcome their addiction (R)
μ	The rate of students leaving all groups

SEIR Model

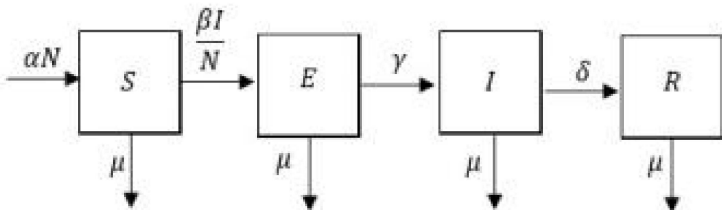


Figure: SEIR Model Schematic for Online Game Addiction

The Equations Governing the Model

Based on the scheme in Figure above, a mathematical model is obtained in a system of differential equations is presented as;

$$\frac{dS}{dt} = \alpha N - \frac{\beta I}{N} S - \mu S$$

$$\frac{dE}{dt} = \frac{\beta I}{N} S - \gamma E - \mu E$$

$$\frac{dI}{dt} = \gamma E - \delta I - \mu I$$

$$\frac{dR}{dt} = \delta I - \mu R$$

Equilibrium Analysis

- Online Game Addiction-Free Equilibrium Point: The addiction-free equilibrium point is obtained when $I = 0$, then

$$S, E, I, R = 1, 0, 0, 0$$

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- Equilibrium Point for Persistent Online Game Addiction: The equilibrium point of online game addiction based on (S^*, E^*, I^*, R^*) is:

$$S^* = \frac{\delta\gamma + \delta\mu + \gamma\mu + \mu^2}{\beta\gamma}, \quad E^* = \frac{(\alpha - \mu)(\delta + \mu)}{\beta\gamma},$$
$$I^* = \frac{\alpha - \mu}{\mu}, \quad R^* = \frac{\delta(\alpha - \mu)}{\mu}$$

The Next Generation Matrix (NGM)

We use the NGM to compute the Basic Reproductive Number (R_0)

The NGM is computed by taking the compartments in the model that are addicted (I) or contribute to the production of new addicts (E)

- New Infections Matrix:

$$\mathcal{F} = \begin{pmatrix} \frac{\beta I}{N} S \\ 0 \end{pmatrix} \text{ or } \mathcal{F}' = \begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix}$$

- Transition Matrix:

$$\mathcal{V} = \begin{pmatrix} -(\gamma + \mu)E \\ \gamma E - (\alpha + \mu)I \end{pmatrix} \text{ or } \mathcal{V}^{-1} = \begin{pmatrix} \frac{-1}{\gamma + \mu} & 0 \\ \frac{-\gamma}{(\delta + \mu)(\gamma + \mu)} & \frac{-1}{\delta + \mu} \end{pmatrix}$$

The NGM Cont.

The product of $\mathcal{F}'$ and $\mathcal{V}^{-1}$ gives us The Next Generation Matrix.

$$\begin{aligned}\mathcal{F}'\mathcal{V}^{-1} &= \begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{-1}{\gamma+\mu} & 0 \\ \frac{-\gamma}{(\delta+\mu)(\gamma+\mu)} & \frac{-1}{\delta+\mu} \end{pmatrix} \\ &= \begin{pmatrix} \frac{-\beta\gamma}{(\delta+\mu)(\gamma+\mu)} & \frac{-\beta}{\delta+\mu} \\ 0 & 0 \end{pmatrix}\end{aligned}$$

Basic Reproductive Number R_0

The basic reproduction number R_0 is determined by calculating the largest eigenvalue in absolute terms.

Thus

$$R_0 = \frac{\beta\gamma}{(\delta + \mu)(\gamma + \mu)}$$

The Model with Optimal Control

From this point forward, we modify the model's control analysis by optimal control which is supported by the Pontryagin Minimum Principle.

Given control variable

$$U = \{u(t) \mid 0 \leq u(t) \leq 1, \forall t \in [0, T]\}$$

where $u(t)$ is the control with guidance and counseling.

The Model with Optimal Control Cont.

The inclusion of the control variable affects the system of equations which describes the model.

Modification of Equations

The new set of equations which describes the model is given as;

$$\frac{dS}{dt} = \alpha N - \frac{\beta I}{N} S - \mu S$$

$$\frac{dE}{dt} = \frac{\beta I}{N} S - (1 - u(t))\gamma E - \mu E$$

$$\frac{dI}{dt} = (1 - u(t))\gamma E - \delta I - \mu I$$

$$\frac{dR}{dt} = \delta I - \mu R$$

Pontryagin Minimum Principle

The objective function of the system of equations is presented as

$$J(u) = \int_0^T \left(E(t) + \frac{1}{2}CU^2 \right) dt$$

with

$$U = \{u(t) \mid 0 \leq u(t) \leq 1, \forall t \in [0, T]\},$$

Where T is the end time and C is the cost factor of guidance and counseling.

Pontryagin Minimum Principle

- The Lagrangian equation of the optimal control is given as:

$$\mathcal{L}(I, U) = E(t) + \frac{1}{2}CU^2$$

- Lagrangian Multiplier:

$$\lambda^T = (\lambda_1 \quad \lambda_2 \quad \lambda_3 \quad \lambda_4)$$

The summation of the Lagrangian equation and the constraint functions multiplied by the Lagrange multiplier gives us the Hamiltonian function.

Pontryagin Minimum Principle

The Hamiltonian function is presented as:

$$\mathcal{H} = \{S(t), E(t), I(t), R(t), u(t), \lambda_1(t), \lambda_2(t), \lambda_3(t)\lambda_4(t)\}$$

$$\begin{aligned}\mathcal{H} = & E(t) + \frac{1}{2}CU^2 + \lambda_1(t) \left(\alpha N - \frac{\beta I}{N}S - \mu S \right) + \\ & \lambda_2(t) \left(\frac{\beta I}{N}S - (1 - u(t))\gamma E - \mu E \right) + \\ & \lambda_3(t)((1 - u(t))\gamma E - \delta I - \mu I) + \lambda_4(t)(\delta I + \mu R)\end{aligned}$$

Pontryagin Minimum Principle

The stationary conditions

$$\frac{\partial \mathcal{H}}{\partial U} = CU + \lambda_2 \gamma E - \lambda_3 \gamma E = 0$$
$$U^* = \frac{\lambda_2 \gamma E - \lambda_3 \gamma E}{C}$$

must be satisfied in order to obtain the optimal conditions of $\mathcal{H}$.

Based on U^* the range of the control variable is $0 \leq u(t) \leq 1$, while the optimal solution $u(t)$ is presented in

$$U^* = \begin{cases} 0, & \text{if } \frac{\lambda_2 \gamma E - \lambda_3 \gamma E}{C} \leq 0 \\ \frac{\lambda_2 \gamma E - \lambda_3 \gamma E}{C}, & \text{if } 0 \leq \frac{\lambda_2 \gamma E - \lambda_3 \gamma E}{C} \leq 1 \\ 1, & \text{if } \frac{\lambda_2 \gamma E - \lambda_3 \gamma E}{C} \geq 1 \end{cases}$$

The optimal solution can be written as Equation

$$U^* = \min \left\{ 1, \max \left(0, \frac{\lambda_2 \gamma E - \lambda_3 \gamma E}{c} \right) \right\}$$

Solving for Optimal Conditions

State Equations

$$\frac{\partial \mathcal{H}}{\partial \lambda_1} = \alpha - \beta IS - \mu S$$

$$\frac{\partial \mathcal{H}}{\partial \lambda_2} = \beta IS - \lambda E + \lambda EU - \mu E$$

$$\frac{\partial \mathcal{H}}{\partial \lambda_3} = \lambda E - \lambda EU - \delta I - \mu I$$

$$\frac{\partial \mathcal{H}}{\partial \lambda_3} = \delta I - \mu R$$

Co-State Equation

$$\frac{\partial \mathcal{H}}{\partial S} = \lambda_1 \beta I + \lambda_1 \mu - \lambda_2 \beta I$$

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial E} &= 1 + \lambda_2 \gamma - \lambda_2 \gamma U + \lambda_2 \mu \\ &\quad + \lambda_3 \gamma + \lambda_2 \gamma U \end{aligned}$$

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial I} &= -\lambda_1 \beta S - \lambda_2 \beta S + \lambda_3 \delta \\ &\quad + \lambda_3 \mu + \lambda_4 \delta \end{aligned}$$

$$\frac{\partial \mathcal{H}}{\partial R} = -\lambda_4 \mu$$

R_0 after the Optimal Control

With initial condition

$$\lambda_1(T) = \lambda_2(T) = \lambda_3(T) = \lambda_4(T) = 0$$

The basic reproduction number formula R_0 , which is obtained after applying the control using the Generation Matrix method is presented as;

$$R_0 = (1 - u^*) \frac{\beta\gamma}{(\delta + \mu)(\gamma + \mu)}$$

Result of Numerical Simulation

Variable & Parameters	Initial Value
S_0	0.62
E_0	0.22
I_0	0.023
R_0	0.05
μ	0.097
α	0.438
β	0.102
γ	0.051
δ	0.0612
C	125

Results of Numerical Simulation

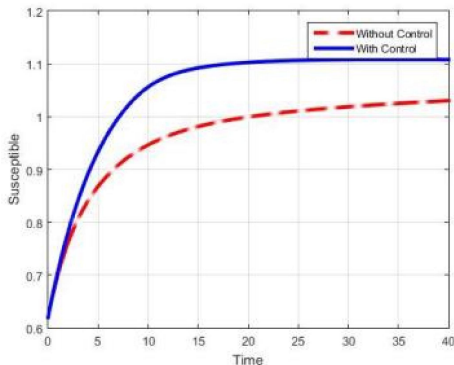


Figure: This figure shows a significant difference between students who are prone to online game addiction before and after intervention with guidance and counseling. Also, there is an increase in the number of students who are prone to addiction after intervention with guidance and counseling.

Results of Numerical Simulation

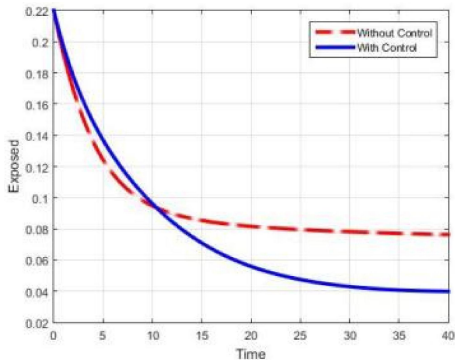


Figure: This figure shows a significant difference in the population of students who started playing online games before and after the intervention with guidance and counseling.

Results of Numerical Simulation

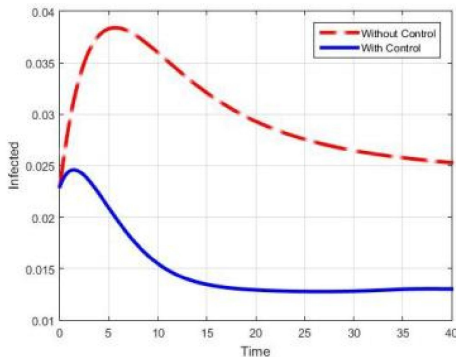


Figure: Shows a significant difference in the population of students addicted to online games before and after the intervention through guidance and counseling. Also, there is a decrease in the number of students addicted to online games after the intervention through guidance and counseling.

Results of Numerical Simulation

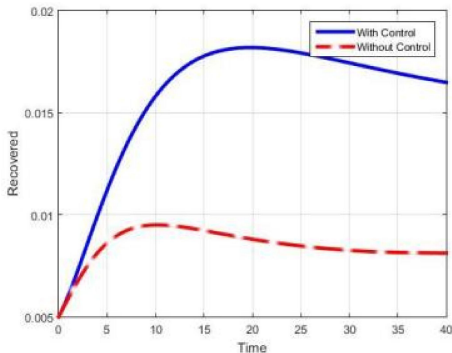


Figure: The study shows a significant shift in students' online gaming habits following guidance and counseling interventions. There was a decrease in the number of students addicted to online games and an increase in those who stopped playing. Numerical simulations confirm that guidance and counseling effectively influenced these changes. Overall, the results demonstrate the positive impact of these interventions in addressing online gaming addiction.

Similar Models at a Glance

Aspect	Article 1	Article 2	Article 3
Model	SEIR	SEIRS	SEIRS
Control Strategy	Guidance and counseling	None	Parental supervision
Target Population	Junior high school students	University mathematics students	Junior high school students
Basic Reproduction Number (R_0)	0.2221 $\rightarrow$ 0.1342 (after control)	0.033 (low transmission)	1.944 (high) $\rightarrow$ 0.556 (after control)
Simulation Tool	Runge-Kutta method in Maple	Maple	Matlab
Outcome Focus	Effectiveness of counseling	Long-term addiction stabilization	Parental supervision efficacy

Figure: Comparison of Online Game Addiction Models

Conclusion

- The SEIR model & optimal control strategies effectively reduce the prevalence of online game addiction
- $R_0 = 0.2221$ became $R_0 = 0.1342$
- Online game addiction can be overcome
- Optimal control through guidance and counseling are crucial interventions to reduce the number of students addicted to online games.

Thanks



- Abdul Saman, Syafruddin Side, Muhammad Isbar Pratama, and Wahidah Sanusi, Optimal Control of the SEIR Model of Online Game Addiction Using Guidance and Counseling, IAENG Letters, 2022.
- Syafruddin Side, Miftahul Jannah, Abdul Saman, Muhammad Isbar Pratama, Wahidah Sanusi, Optimal Control and Analysis of the SEIRS Model on the Problem of Online Game Addiction: A Case Study Among Class VIII Students of the State Junior High Schools in Makassar City, International Journal of Applied Mathematics, 2024.
- Anwar, SEIRS Model Analysis for Online Game Addiction Problem of Mathematics Students, Journal of Physics, 2021.